



High-Dimensional Statistics

Fall 2025

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Assignment 1

Instructions

- **Due.** November 9, 2025
- **Scoring.** The assignment contains **120 points** in total, but **100 points** are sufficient to receive full credit. Extra points will not be carried over to other assignments.
- **Late policy.** A maximum of three assignments may be submitted late, each with a delay of up to 3 days. Submissions beyond this limit or after the allowed delay will not be accepted.
- **Collaboration.** If you discuss concepts with classmates, list the collaborators.
- State assumptions, cite any less well-known lemmas or results you use, and keep proofs concise.

Problem 1. Classical Statistics

[10 points]

Assume that $T = t(X)$ is an estimator with expectation $\psi(\theta)$ (based on the observations X), i.e.,

$$\mathbb{E}(T) = \psi(\theta).$$

Prove that, for all θ ,

$$\text{var}(t(X)) \geq \frac{[\psi'(\theta)]^2}{I(\theta)},$$

where $I(\theta)$ is the Fisher information, defined as

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right],$$

where $f(X; \theta)$ is the probability density (or mass) function of X given the parameter θ .

Problem 2.

[10 points]

Let $X \in \mathbb{R}^d$ be a sub-Gaussian random vector with variance proxy σ^2 . Then

$$\mathbb{E} \left[\max_{\theta \in \mathcal{B}_2} \theta^\top X \right] = \mathbb{E} \left[\max_{\theta \in \mathcal{B}_2} |\theta^\top X| \right] \leq 4\sigma\sqrt{d}.$$

Moreover, for any $\delta > 0$, with probability $1 - \delta$, it holds

$$\max_{\theta \in \mathcal{B}_2} \theta^\top X = \max_{\theta \in \mathcal{B}_2} |\theta^\top X| \leq 4\sigma\sqrt{d} + 2\sigma\sqrt{2\log(1/\delta)}.$$

Problem 3.

[10 points]

Let X be a mean-zero random variable such that $|X| \leq K$. Prove the following bound on the MGF of X :

$$\mathbb{E} \left[e^{\lambda X} \right] \leq \exp \left(g(\lambda) \mathbb{E}[X^2] \right), \quad \text{where} \quad g(\lambda) = \frac{\lambda^2/2}{1 - |\lambda|K/3},$$

provided that $|\lambda| < 3/K$.

Problem 4.

[10 points]

Let $X_1, \dots, X_N$ be independent sub-gaussian random variables with zero means and unit variances, and let $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Prove that for every $p \in [2, \infty)$ we have

$$\left(\sum_{i=1}^N a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^N a_i X_i \right\|_{L^p} \leq CK \sqrt{p} \left(\sum_{i=1}^N a_i^2 \right)^{1/2}$$

where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.

Problem 5. Expectation of maximum

[10 points]

Let $X_1, \dots, X_n$ be a sequence of σ -subgaussian random variables (possibly dependent) and

$$Z = \max_{t \in [n]} X_t, \quad [n] := \{1, \dots, n\}.$$

Prove that

- (a) $\mathbb{E}[Z] \leq \sqrt{2\sigma^2 \log n}$.
- (b) $\mathbb{P}\left(Z \geq \sqrt{2\sigma^2 \log\left(\frac{n}{\delta}\right)}\right) \leq \delta$ for any $\delta \in (0, 1)$.

Problem 6. Sample mean concentration for categorical distributions

[10 points]

Let $X_1, \dots, X_n$ be an independent and identically distributed sequence taking values in $[m]$. For $i \in [m]$, let

$$p(i) = \mathbb{P}(X_1 = i) \quad \text{and} \quad \hat{p}(i) = \frac{1}{n} \sum_{t=1}^n \mathbb{I}\{X_t = i\}.$$

Show that for any $\delta \in (0, 1)$,

$$\mathbb{P}\left(\|p - \hat{p}\|_1 \geq \sqrt{\frac{2 \left[\log\left(\frac{1}{\delta}\right) + m \log(2)\right]}{n}}\right) \leq \delta.$$

Problem 7. Subgaussianity of Bernoulli distributions

[10 points]

Let X be a random variable with Bernoulli distribution with mean p . That is, $X \sim \text{Bern}(p)$: $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$.

- (a) Show that X is $1/2$ -subgaussian for all p .
- (b) Let $Q : [0, 1] \rightarrow [0, 1/2]$ be the function

$$Q(p) = \sqrt{\frac{1 - 2p}{2 \ln\left(\frac{1-p}{p}\right)}},$$

where undefined points are defined in terms of their limits. Show that X is $Q(p)$ -subgaussian.

- (c) The subgaussianity constant of a random variable X is the smallest value of σ such that X is σ -subgaussian. Show that the subgaussianity constant of $X \sim \mathcal{B}(p)$ is $Q(p)$.

(d) Show that for $\lambda \geq 0$ and $p \geq \frac{1}{2}$,

$$\mathbb{E}[\exp(\lambda X)] \leq \exp\left(\frac{p(1-p)\lambda^2}{2}\right).$$

Think of how these inequalities are used for bounding tails. What do you conclude?

Problem 8. PAC Bounds for Best-Arm Identification

[50 points]

You are given a stochastic n -armed bandit with arms $A = \{a_1, \dots, a_n\}$. Pulling arm a yields a reward $R(a) \in [0, 1]$ with unknown mean $p_a = \mathbb{E}[R(a)]$. Let a^* be a best arm with mean $p^* = \max_a p_a$.

An arm a is ε -optimal if $p_a \geq p^* - \varepsilon$. An algorithm is (ε, δ) -PAC if, upon termination, it outputs an ε -optimal arm with probability at least $1 - \delta$.

We try to find a near-optimal arm with fewer samples. We'll look at four algorithms, each improving on the previous one.

1. Naive Sampling

The simplest idea is to sample each arm the same number of times, compute the average reward, and pick the arm with the highest empirical mean.

Algorithm 1 Naive(ε, δ)

- 1: **for** every arm $a \in A$ **do**
 - 2: Sample it $\ell = \frac{4}{\varepsilon^2} \log\left(\frac{2n}{\delta}\right)$ times.
 - 3: **end for**
 - 4: Let $\hat{p}_a$ be the average reward of arm a .
 - 5: **output** $a' = \arg \max_{a \in A} \{\hat{p}_a\}$.
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This uniform-sampling rule ignores the fact that some arms are clearly worse early on. You'll see shortly that while it is correct, it can be wasteful.

2. Successive Elimination (Known Gaps)

Suppose, you know in advance the "gaps"

$$\Delta_i = p_1 - p_i,$$

where $p_1 \geq p_2 \geq \dots \geq p_n$. Then you can plan to eliminate arms one by one in phases, sampling less as you get closer to the best.

Algorithm 2 Successive Elimination Known Biases(δ)

- 1: Set $S = A$; $t_i = \left\lceil \frac{4}{\Delta_i^2} \log\left(\frac{n}{\delta}\right) \right\rceil$; and $t_{n+1} = 0$.
 - 2: Set for every arm a : $\hat{p}_a = 0$.
 - 3: **for** $i = 0$ to $n - 2$ **do**
 - 4: (a) Sample every arm $a \in S$ for $t_{n-i} - t_{n-i+1}$ times.
 - 5: (b) Let $\hat{p}_a$ be the average reward of arm a (in all rounds).
 - 6: (c) Set $S = S \setminus a_{\min}$, where $a_{\min} = \arg \min_{a \in S} \{\hat{p}_a\}$.
 - 7: **end for**
 - 8: **output** S .
-

This algorithm knows the difficulty of distinguishing arms (Δ_i) ahead of time. It adapts sampling effort accordingly and achieves a gap-dependent bound.

3. Successive Elimination (Unknown Gaps)

In practice, we don't know the Δ_i 's. The following variant adaptively estimates and eliminates arms using a confidence radius that shrinks over time.

Algorithm 3 Successive Elimination(δ)

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1: Set  $t = 1$  and  $S = A$ .
2: Set for every arm  $a$ :  $\hat{p}_a^1 = 0$ .
3: repeat
4:   Sample every arm  $a \in S$  once and let  $\hat{p}_a^t$  be the average reward of arm  $a$  by time  $t$ .
5:   Let  $\hat{p}_{\max}^t = \max_{a \in S} \hat{p}_a^t$  and  $\alpha_t = \sqrt{\frac{\log(cnt^2/\delta)}{t}}$ .
6:   For every arm  $a \in S$  such that  $\hat{p}_{\max}^t - \hat{p}_a^t \geq 2\alpha_t$ , set  $S \leftarrow S \setminus \{a\}$ .
7:    $t \leftarrow t + 1$ .
8: until  $|S| \leq 1$ 
9: output  $S$ .
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By using tighter confidence intervals as t increases, this algorithm never eliminates the best arm (with high probability) and gradually prunes away suboptimal ones.

4. Median Elimination

The last algorithm uses a clever “median trick.” In each phase, it samples all surviving arms enough to estimate their means accurately, then discards the bottom half. This simple idea reduces the dependence on $\log n$ in the sample complexity.

Algorithm 4 Median Elimination(ϵ, δ)

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1: Set  $S = A$ .
2:  $\epsilon_1 = \epsilon/4$ ,  $\delta_1 = \delta/2$ ,  $\ell = 1$ .
3: while True do
4:   Sample every arm  $a \in S$  for  $\frac{1}{(\epsilon_\ell/2)^2} \ln\left(\frac{3}{\delta_\ell}\right)$  times, and let  $\hat{p}_a^\ell$  denote its empirical value.
5:   Find the median of  $\hat{p}_a^\ell$ , denoted by  $m_\ell$ .
6:    $S_{\ell+1} = S_\ell \setminus \{a : \hat{p}_a^\ell < m_\ell\}$ .
7:   if  $|S_\ell| = 1$  then
8:     output  $S_\ell$ .
9:     break.
10:  else
11:     $\epsilon_{\ell+1} = \frac{3}{4}\epsilon_\ell$ ,  $\delta_{\ell+1} = \delta_\ell/2$ ,  $\ell = \ell + 1$ ; Go to step 3.
12:  end if
13: end while
```

Answer the following questions based on the given algorithms.

- (a) Prove that the algorithm NAIVE(ϵ, δ) is an (ϵ, δ) -PAC algorithm with arm sample complexity $O\left(\frac{n}{\epsilon^2} \log\left(\frac{n}{\delta}\right)\right)$.
- (b) Prove that the SUCCESSIVE ELIMINATION WITH KNOWN BIASES algorithm is a $(0, \delta)$ -PAC algorithm and its arm sample complexity is

$$O\left(\log\left(\frac{n}{\delta}\right) \sum_{i=2}^n \frac{1}{\Delta_i^2}\right).$$

- (c) Prove that the SUCCESSIVE ELIMINATION algorithm is a $(0, \delta)$ -PAC algorithm, and with

probability at least $1 - \delta$ its arm sample complexity is bounded by

$$O\left(\sum_{i=2}^n \frac{\log\left(\frac{n}{\delta\Delta_i}\right)}{\Delta_i^2}\right).$$

(d) Prove that

1) For the MEDIAN ELIMINATION(ϵ, δ) algorithm we have

$$\Pr\left[\max_{j \in S_\ell} p_j \leq \max_{i \in S_{\ell+1}} p_i + \epsilon_\ell\right] \geq 1 - \delta_\ell.$$

2) The MEDIAN ELIMINATION(ϵ, δ) algorithm has arm sample complexity

$$O\left(\frac{n}{\epsilon^2} \ln \frac{1}{\delta}\right).$$

3) The MEDIAN ELIMINATION(ϵ, δ) algorithm is an (ϵ, δ) -PAC algorithm and its arm sample complexity is

$$O\left(\frac{n}{\epsilon^2} \ln \frac{1}{\delta}\right).$$